

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$ (1)						
				$K_c = 0.119$ (1)					1	
				$\text{dm}^6 \text{ mol}^{-2}$ (1)		3			1	
				student is correct since $K_c < 1$ / very small (1)			1	4		
		(ii)		student incorrect since equilibrium shifts to left (1)			1			
				system opposes change by taking in heat, favouring endothermic direction (1)	1			2		
	(b)			$n(\text{NH}_3) = 58720$ (1)					1	
				$n(\text{DAP}) = 22020$ (1)					1	
				maximum mass DAP = 2909 (1)		3		3		
	(c)			$pV = nRT$ (1)	1				1	
				$V = \frac{2.54 \times 10^{-3} \times 8.31 \times 393}{101000}$ (1)					1	
				$V = 8.21 \times 10^{-5} \text{ m}^3$ (1)					1	
				$V = 82.1 \text{ cm}^3$ (1)		3		4		
				credit alternative method using molar volume						
				Question 10 total	2	9	2	13	7	0